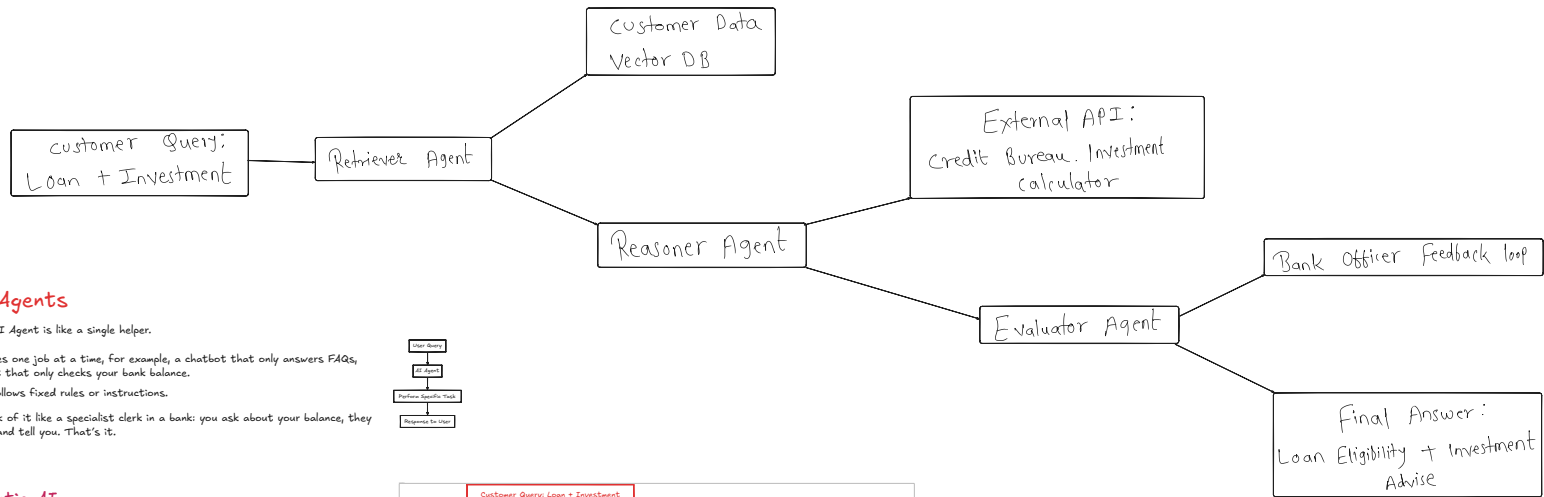




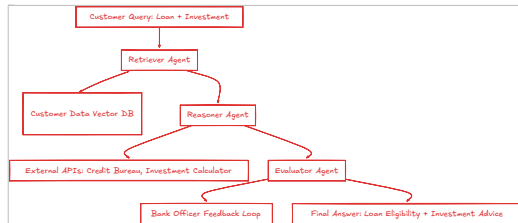
AI Agents

- An AI Agent is like a single helper.
- It does one job at a time, for example, a chatbot that only answers FAQs, or a bot that only checks your bank balance.
- It follows fixed rules or instructions.
- Think of it like a specialist clerk in a bank: you ask about your balance, they check and tell you. That's it.



Agentic AI

- Agentic AI is like a team of helpers working together.
- Each agent has a role (fetch data, reason, check correctness, give final answer).
- They can adapt, collaborate, and use tools to solve bigger problems.
- Think of it like a whole banking team: One clerk fetches your file (Retriever).
- One officer studies your file and calls external systems (Reasoner).
- One manager checks the decision (Evaluator).
- Together they give you a complete answer: loan eligibility + investment advice.



Key Difference

- AI Agent = one worker, one task.
- Agentic AI = a team of workers, each with a role, collaborating to solve complex tasks.
- AI Agents are single-task bots.
- Agentic AI is multi-agent teamwork, orchestrated by frameworks like LangGraph.

Scope	Single task, narrow focus
Autonomy	Limited, rule-based
Decision-Making	Predefined, fixed workflows
Collaboration	Works alone
Example 1	Customer service chatbot (answers FAQs only)
Example 2	Banking bot (checks account balance only)

AI Agents

Agentic AI

- Multi-task, adaptive, broader goals
- High, learns and adapts
- Contextual, situational, dynamic
- Multiple agents coordinate
- Smart home system (optimizes energy with multiple agents)
- Banking workflow (Retriever + Reasoner + Evaluator for loan + investment)

AI Agents are individual software programs that perform specific tasks autonomously. They follow predefined rules or instructions and focus on narrow, single-purpose jobs.

Key Features:

- Single-task, narrow scope
- Limited autonomy, rule-based
- Predefined decision-making
- Works alone without collaboration

Real-World Examples:

- Customer Service Chatbot**
Responds to FAQs like password reset or account balance inquiry.
- Banking Bot**
Automates simple tasks such as checking account balance or sending transaction alerts.
- E-commerce Assistant**
Tracks orders, processes returns, and sends shipping updates.
- Automated Email Responder**
Replies to standard queries with fixed templates.
- Scheduling Bot**
Books meetings or appointments based on predefined rules.

Agentic AI is a system where multiple AI agents collaborate to solve complex tasks.

Each agent has a specialized role such as Retriever, Reasoner, or Evaluator, and together they form an adaptive workflow.

Key Features:

- Multi-task, adaptive decision-making
- Collaboration between agents
- Integration with memory, tools, and human feedback

Real-World Applications:

- Smart Home Energy Optimization**
Agents coordinate occupancy detection, electricity pricing, and device control.
Outcome: reduced bills and efficient energy use.
- Banking Workflow Automation**
Retriever fetches customer data, Reasoner checks credit rules, Evaluator validates compliance.
Outcome: personalized loan eligibility and investment advice.
- Healthcare Assistant**
Agents analyze patient records, suggest treatments, schedule follow-ups, and alert doctors.
Outcome: tailored healthcare recommendations.
- Supply Chain Management**
Agents track inventory, predict demand, optimize logistics, and flag risks.
Outcome: smoother operations and reduced costs.
- Education Tutor System**
Retriever pulls learning materials, Reasoner adapts to student progress, Evaluator checks accuracy.
Outcome: personalized learning paths for students.